Mohamed Mehdi Keteb

PhD student in Statistics

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Research interests : I am interested in sampling problems, Monte Carlo methods (MCMC and unbiased MCMC) and Markov chain theory (spectral gaps, couplings and ergodicity). Now, I am focusing on estimation of log-normalizing constant, path sampling, tempering path and coupling.

Education

PhD student in Statistics and Probability at ESSEC Business School *Cergy, FR*

Department of Information Systems, Data Analytics & Operations *2025 - 2028*

- **Supervisor :** Pierre E. Jacob (`https://pierrejacob.org/`).
- **Context :** I am part of the ERC Consolidator grants project of Pierre E. Jacob titled UMCMC Markov Chain Monte Carlo using couplings toward scalable statistical inference (`https://www.insmi.cnrs.fr/fr/personne/pierre-jacob`).
- **Title :** Scalability, efficiency and robustness of unbiased Markov chain Monte Carlo (`https://theses.fr/s422329`) .
- **My current Project :** I am working on an unbiased MCMC estimator based on coupling and path sampling (also known as thermodynamic integration) to produce an efficient estimation of the Log normalizing constant and provides confidence intervals via the CLT, leveraging parallel processing.

Université Paris-Saclay - Institute of Mathematics of Orsay *Orsay, FR*

M2 Mathematics of rondomness - Statistics and Machine learning *2024 - 2025*

- **Modules :** Statistical learning (S.Arlot), Concentration inequalities (P.Massart), High dimensional statistics (C.Giraud), High dimensional probability (M.Lerasle), Non-parametric estimation (V.Divol), Non-parametric Bayesian methods (I.Castillo), Topological Data Analysis (F.Chazal), Stochastic approximation applied to reinforcement learning (W.Hachem & P.Bianchi)

ENSAE Paris - Polytechnic Institute of Paris *Paris, FR*

Master of Science in applied mathematics : statistics, machine learning and data science *2022 - 2025*

- **3rd year modules - 4/4 GPA:** Statistical Optimal transport (A.Stromme), Advanced statistics (V.E.Brunel), Bayesian statistics, Advanced optimization (V.Perchet), Generative modeling (A.Korba), NLP, Matrix Data Analysis (M.Lerasle), Algorithm Design (P.Loiseau), Ethics and responsibility in data science.
- **2nd year modules - 4/4 GPA :** Statistics (A.Dalalyan), Stochastic process (N.Chopin), Monte carlo (N.Chopin), Theoritical foundations of Machine learning (V.Perchet), Time series
- **1st year modules - 4/4 GPA :** Functional and convex analysis (L.Decreusefond), Probability theory (V.E.Brunel), Statistics (M.Lerasle), Convex optimization (G.Lecué), Theory of measurement (C.Butucea)

Lycée Thiers *Marseille, FR*

Preparatory class for the French engineering schools - MPSI-MP* *2020 - 2022*

- **Relevant courses :** Mathematics (Algebra, Topology, Analysis, Probability), Theoretical Physics, Computer Science (Python, SQL)

Research experiences

Department of Information Systems, Data Analytics & Operations - ESSEC Business School *Cergy, FR*

Research assistant in probability and statistics *Jun 2025 - Sep 2025*

- Explore the possibility of coupling the Elliptical Slice Sampling algorithm to analyze contraction properties or meeting times, with the goal of constructing unbiased estimators.
- Preparatory internship for the PhD, conducted under the same supervisor.
- Supervised by Pierre E. Jacob

Department of mathematics - King's college London *London, UK*

Research assistant in probability and statistics *Jun 2024 - Sep 2024*

- Convergence analysis of the particle swarm optimization algorithm (PSO) (which is a stochastic optimization algorithm) using Markov chain theory
- Supervised by Francesca Romana Crucinio

Centre for research in economics and statistics (CREST) - Polytechnic Institute of Paris *Paris, FR*

Research assistant in probability and statistics *Jun 2023 - Jul 2023*

- Research subject: "Wasserstein distance in the case of mixing measures applied to Topic Modeling" based on the work of Florentina Bunea (Cornell University)
- Supervised by Christina Butucea

Posters / Seminars / Reading Groups / Workshops

Posters :

Poster at The 2026 ISBA World Meeting *Nagoya, Japan*

The purpose of the meeting is to bring together the diverse international community of investigators in statistics who develop and use Bayesian methods *Upcoming*

- I will be presenting my joint work with Pierre Jacob on Unbiased estimation of log-normalizing constants
- Poster title : Coupling-based Unbiased Estimation of the Log Normalizing Constant via Path Sampling

Poster at ISBA Satellite Meeting 2026 *Tokyo, Japan*

Information Geometry, Privacy and Monte Carlo *Upcoming*

- I will be presenting my joint work with Pierre Jacob on Unbiased estimation of log-normalizing constants
- Poster title : Coupling-based Unbiased Estimation of the Log Normalizing Constant via Path Sampling

Reading Groups :

Reading group at Parisanté Campus *Paris, FR*

Joint reading group with the teams of Nicolas Chopin, Pierre E. Jacob, and Christian Robert *Ongoing*

- Log-Concave Sampling by Sinho Chewi
- Explicit convergence bounds for Metropolis Markov chains by Andrieu et al.
- For more details: `https://pierrejacob.quarto.pub/psc-friday-afternoon-reading-group/`

Seminars :

Seminar of The Department of Information Systems, Data Analytics & Operations - ESSEC Business School *Cergy, FR*

Seminar for Phd student on sampling, Bayesian statistics, probability, econometrics and more *Ongoing*

- The speakers : Takamitsu Kurita (Kyoto Sangyo University), Peter Orbanz (Gatsby), David Rossell (Universitat Pompeu Fabra) and more

Seminar of Université Paris-Saclay - Institute of Mathematics of Orsay *Orsay, FR*

Seminar at the Orsay Mathematics Laboratory as part of my Master's program. *Oct 2024 - Jan 2025*

- The speakers were : Gilles Stoltz, Vianney Perchet, Claire Lacour, Nicolas Chopin, Guillaume Lecué, Anne Sabourin, Aurélie Fisher, Zacharie Naulet, Sylvain Le corff, Erwan Scornet, Ismaël Castillo

Workshops :

Workshop on MCMC *Zürich, Switzerland*

Scalable MCMC Sampling - Institute for Mathematical Research, ETH Zürich *Upcoming*

International workshop on diffusions in machine learning *London, UK*

Isaac Newton Institute for Mathematical Sciences *15th Jul 2024 - 19th Jul 2024*

- I participated in this workshop to understand more about diffusion and sampling

Projects

Theoretical Statistics Project *Paris, FR*

ENSAE Paris *May 2024 - Jun 2025*

- Overview of Gaussian laws in normed vector spaces, possibly infinite-dimensional, and their connection to Gaussian processes. Discuss central limit theorems for i.i.d. variables, highlighting the role of type and cotype in Banach spaces. Examine the link between these results, Donsker classes, and the asymptotic normality of empirical processes.

NLP Project *Paris, FR*

ENSAE Paris *Apr 2025 - May 2025*

- Compare different classifiers with various embeddings across multiple datasets to evaluate the robustness of the models for the fake news classification task.

Optimal Transport and Deep learning Project *Paris, FR*

ENSAE Paris *Dec 2024 - Jan 2025*

- Create a new model based on the Wasserstein Distance to perform the task of image retrieval
- Prove Varadhan's formula in the Euclidean case.
- Literature review of the well-known article on the Convolutional Wasserstein distance

Bayesian Statistics Project *Paris, FR*
ENSAE Paris *Dec 2024 - Jan 2025*
- Estimate the crash rate of an airline using various priors, particularly a log-normal prior, with the Metropolis-Hastings algorithm in order to make predictions

Sequential Monte Carlo Project *Paris, FR*
ENSAE Paris *Nov 2023 - May 2024*
- The project focused on applying an ABC-SMC method to sample from an alpha-stable model

Machine learning forecasting applied to cryptocurrency prices in Python *Paris, FR*
ENSAE Paris *Oct 2023 - Dec 2023*
- Model used : SVM, Neural Network, Linear Regression and ARIMA

Delivery network optimization project in Python *Paris, FR*
ENSAE Paris *Jan 2023 - Apr 2023*
- Research and optimization of the route connecting two cities in a delivery network (graph)
- Optimization of agent acquisition to maximize profit generated by a delivery network (knapsack problem)

Skills and Hobbies

Languages : French (native), English (Proficient)
Coding : Python, R, LaTeX, Git/Github
Hobbies : Walking, Fishing, Weight Training, Hiking, Cycling, Backpacking